

Vendor Performance Analysis Report

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Role: Data Analyst Project

Tools Used: SQL | Python | Power BI

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1. Executive Summary

This project analyzes vendor performance to evaluate revenue, cost, profitability, and operational efficiency. The objective is to transform raw transactional data into meaningful insights using SQL, Python, and Power BI.

The dashboard highlights top-performing vendors, identifies cost-heavy vendors, and visualizes how revenue converts into profit using a waterfall chart.

2. Problem Statement

Organizations often struggle to evaluate vendor efficiency due to large and unstructured datasets. Without proper analysis and visualization, it becomes difficult to:

- Identify high-performing vendors
- Detect cost-heavy vendors
- Understand profitability trends

This project addresses these challenges through structured analysis and an interactive dashboard.

3. Objectives

- Analyze total revenue, cost, and profit
- Identify top 10 vendors by revenue
- Compare profit vs cost across vendors
- Evaluate vendor profitability (margin)
- Visualize profit breakdown using waterfall

4. Dataset Overview

The dataset includes:

- Vendor Number (Supplier ID)
- Inventory ID

- Sales Quantity
- Sales Revenue
- Purchase Price
- Inventory Levels

This enables vendor-level performance analysis.

5. Tools & Technologies

- **SQL (MySQL):** Data extraction & querying
- **Python (Pandas):** Data validation & preprocessing
- **Power BI:** Dashboard & visualization
- **CSV Files:** Data storage

6. Data Preparation

- Cleaned missing and inconsistent data
- Converted numeric fields and ensured correct data types
- Joined sales and purchase datasets
- Created calculated measures (Revenue, Cost, Profit, Margin)
- Applied Top 10 filtering for consistency

7. Key Performance Indicators (KPIs)

- **Total Revenue:** 452.06M
- **Total Cost:** 308.90M
- **Total Profit:** 136.68M
- **Profit Margin:** 30.23%

8. SQL Analysis

SQL was used to extract, join, and analyze vendor-level data efficiently.

Key Queries Performed:

- Revenue calculation
- Cost and profit computation
- Vendor-wise performance

- Top 10 vendors by revenue
- Inventory turnover analysis

Sample Queries

-- KPI Calculation

```
SELECT
    s.brand,
    SUM(s.salesdollars) AS revenue,
    SUM(p.purchaseprice * s.salesquantity) AS cost,
    SUM(s.salesdollars - (p.purchaseprice * s.salesquantity)) AS profit
FROM sales s
LEFT JOIN purchases p
ON s.inventoryid = p.inventoryid
GROUP BY s.brand;
```

-- Top Vendors by Revenue

```
SELECT
    s.vendornumber,
    SUM(s.salesdollars) AS revenue
FROM sales s
GROUP BY s.vendornumber
ORDER BY revenue DESC
LIMIT 10;
```

9. Python Analysis

Python was used for data validation and KPI verification.

Tasks Performed:

- Data cleaning using Pandas
- Merging datasets
- Calculating cost and profit
- Validating KPI values

Sample Code

```
import pandas as pd
```

```
sales = pd.read_csv("sales.csv")
```

```
purchases = pd.read_csv("purchases.csv")
```

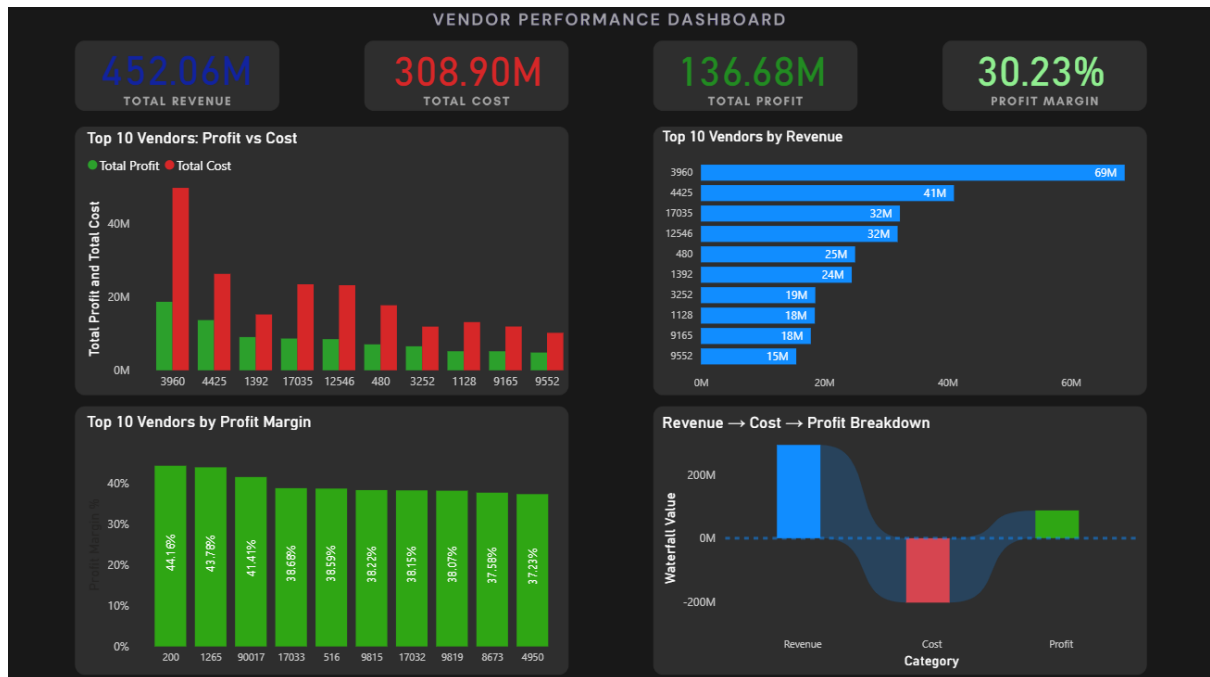
```
df = pd.merge(sales, purchases, on="inventoryid", how="left")
```

```
df["cost"] = df["purchaseprice"] * df["salesquantity"]
df["profit"] = df["salesdollars"] - df["cost"]
```

```
total_revenue = df["salesdollars"].sum()
total_profit = df["profit"].sum()
```

10. Power BI Dashboard

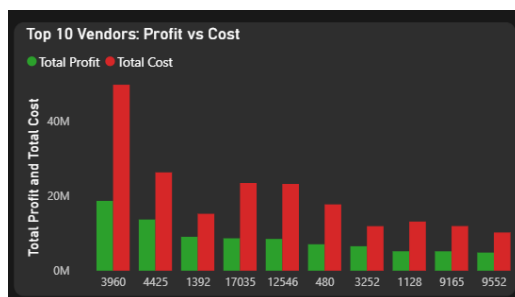
Figure 1: Vendor Performance Dashboard



The dashboard provides a high-level overview of vendor performance, including key KPIs and visual insights.

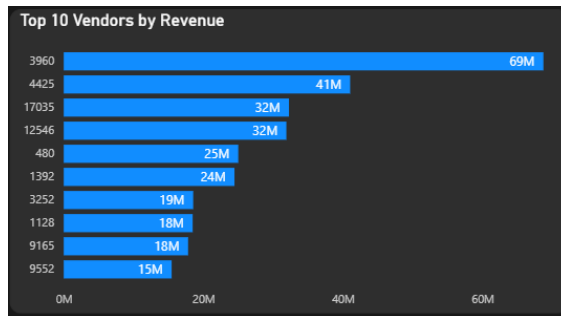
11. Visual Analysis

Figure 2: Top 10 Vendors – Profit vs Cost



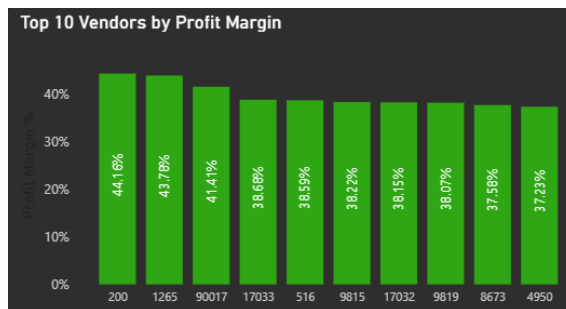
- Compares cost vs profit
- Identifies cost-heavy vendors

Figure 3: Top 10 Vendors by Revenue



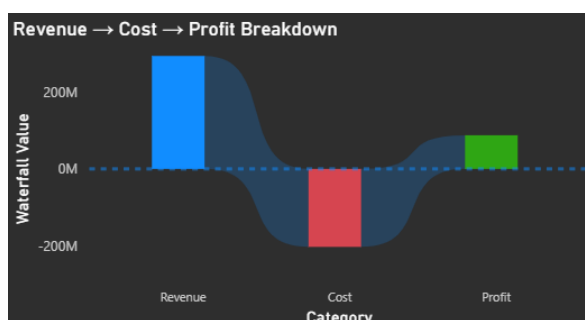
- Highlights highest revenue vendors
- Shows revenue concentration

Figure 4: Top 10 Vendors by Profit Margin



- Identifies most efficient vendors
- Highlights high-margin performers

Figure 5: Revenue → Cost → Profit Breakdown



- Shows revenue transformation into profit
- Clearly highlights cost impact

12. Key Insights

- A small group of vendors contributes most revenue
- High costs significantly reduce profitability
- Some vendors generate high revenue but low margin
- Profit margin is stable (~30%)
- Vendor performance varies significantly

13. Business Value

This dashboard helps businesses:

- Identify high-performing vendors
- Optimize cost-heavy vendor relationships
- Improve profit margins
- Support data-driven decision-making

14. Conclusion

This project demonstrates how SQL, Python, and Power BI can be integrated to analyze vendor performance effectively. The dashboard provides clear and actionable insights for business optimization.

15. Recommendations

- Focus on high-margin vendors
- Reduce dependency on high-cost vendors
- Optimize procurement strategies
- Monitor vendor performance regularly

16. Future Scope

- Vendor segmentation
- Predictive analytics
- Inventory optimization